

Member Contributions & Manual Testing Tasks

Overall Contribution Split

Member	Role	Contribution Focus	Assigned %
Member A (Nguyễn Đình Minh Huy)	Engine & Infrastructure	Game Loop, Physics Engine, Collision Systems, Level/World Systems, Serialization, Level Editor, Replay/Rewind, Multiplayer AI (Shadow Mario, CPU opponent), Advanced Systems.	50%
Member B (Trần Gia Huy)	Entities & Gameplay	Player States & Power-ups, Enemy AI & Bosses, Items & Blocks, Game Flow/UI, Audio, Controls, Visual Effects, Accessibility, Polish & Game Feel.	50%

Manual Testing Task Division

Every one of the 97 features in `features_list.md` is assigned below to the member whose domain it falls into, grouped by that document's own 16 categories so a tester can work through it category by category with the feature numbers open side by side. This replaces the previous six-bullet-per-member sketch with a checklist that actually covers the full list.

Member A — Engine, Physics, World & Systems Testing

#	Category (features)	Manual test
1	Core Engine & Architecture (1–10)	Play for 10+ minutes across at least two levels; confirm no frame-rate stutter, no dangling ImGui window, and that rebinding a key in Options survives a restart. Open the debug console (<code>`</code>) and run every one of <code>help/give/lives/tp/god/spawn/difficulty/level/progress/replay/clear</code> at least once.
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#	Category (features)	Manual test
	Physics & Collision (11–16)	Deliberately test edge cases: jump into a wall corner, walk onto ice (1-2) and a conveyor (1-3) back to back, and stomp an enemy at the exact frame you also touch a block from below. Confirm no jitter or tunnelling.
3	Advanced Movement (17–22)	Wall-jump off a shaft, ground-pound from a Mega state and a normal state, crouch-slide under a 1-tile gap, and chain 3+ stomps without touching ground to confirm the combo counter increments and resets correctly on landing-idle or damage.
4	Levels & World (60–64, 96–97)	Play all 3 main worlds + Bonus Stage to their flagpole, enter and exit all 3 sub-vaults through their pipes, confirm the World Map unlocks each node in order and shows the right star-coin pips, then open the Level Editor (F1), place an entity, save, and reload it. From Main Menu → Procedural Level → Play Endless, run right for at least 2 minutes: confirm the distance counter climbs in the world-label HUD field, at least one chunk boundary is crossed with no visible break in the terrain, and dying shows "WORLD ENDLESS - Nm" with a recorded high score. Use "Generate & Play" 5+ times across different themes/difficulties and confirm every generated level is completable start to flagpole (the solvability check should mean you never see an impossible gap).
5	Save/Load & Persistence (75–78)	Save to a slot mid-level, quit, and reload it — confirm position, lives, coins and score match. Record a run's high score and confirm it appears in Options → Records.
6	Two-Player & AI Opponents (79–82)	Start a Versus Human match and a Versus CPU match (try both a Speedrunner and a Hunter profile) and a Shadow Chase match; confirm Player 2 controls respond independently and Shadow Mario visibly replays your own prior path.
7	Advanced Systems (89–95)	Hold the rewind key mid-jump and confirm the game rewinds cleanly; save and play back a replay from the console; clear the campaign once and confirm New Game+ starts with faster enemies; run the Daily Challenge twice on the same day and confirm the level is identical both times; toggle Colorblind Mode and confirm the minimap recolors.

Member B — Entities, Gameplay & Polish Testing

#	Category (features)	Manual test
1	Characters & Player States (23–31)	Play a full level as each of Mario, Luigi, Toad and Peach, noticing the speed/jump differences. Take every power-up in sequence (Small →

#	Category (features)	Manual test
		Super → Fire → Cape, and separately Mini and Mega) and confirm the hitbox and sprite change each time; take a Star while in Fire form and confirm you return to Fire form when it ends.
8	Enemies & AI (32–41)	Encounter every enemy type at least once (place the editor-only ones — Paratroopa, Boo, Thwomp, Lakitu, Bullet Bill, Chain Chomp — via the Level Editor if they are not in your current playthrough) and confirm each one's movement strategy matches its description (patrol, chase, fly, timer emergence, tethered lunge).
9	Bosses (42–46)	Fight Boom Boom in 1-2 and Bowser in 1-3 to completion. Confirm the boss health bar only appears during the fight, the arena will not let you leave or reach the flagpole until the boss is dead, and each boss's phase 2/3 attack pattern actually changes.
10	Items & Blocks (47–59)	Trigger every item at least once (all 7 power-ups, a POW Block, a P-Switch, a Trampoline) and every block type (Question, Brick, a Moving Platform, a Falling Platform, ice and conveyor tiles, a Warp Pipe). Finish a level and confirm the castle's flag rises and you visibly walk up to its door before the summary screen appears.
11	Game Flow & UI (65–71)	Walk the full Main Menu → Pause → Options → Records loop. Toggle the Minimap (Tab) mid-level, open the Statistics screen, and die on purpose to confirm the death tally and instant "Retry Level" both work.
12	Audio (72–74)	With sound on, confirm each listed SFX actually plays for its trigger (jump, coin, stomp, power-up, damage, fireball, death, flagpole, pipe, ground pound, P-Switch, achievement) and that the music changes between the menu, world map, a level, a boss fight and Game Over. Move both volume sliders and confirm the change is audible.
13	Controls (83–84)	Rebind at least 3 actions to new keys in Options, confirm they work in-level, then use "Reset Controls" and confirm the defaults return.
14	Visual Effects (85–88)	Break a brick, collect a coin, stomp an enemy with a combo active, and take damage — confirm a particle burst, the correct death animation, and the invincibility flash all appear as described.
—	Accessibility (95, cross-referenced with Member A's #7)	Confirm the colourblind palette swap is visible on the minimap and any debug outlays without needing the debug console.

Cross-cutting: play at least one full session together

However the two domains above are split, finish with one full co-op or versus session played together end to end (Main Menu → a full level → Victory → World Map → a second level → Game Over or campaign clear), so at least one manual pass exercises the transitions between every system in this table rather than each system in isolation.